The Image–Text Information Gap in Medical MLLMs

Motivation, formalization, mitigation, and theory

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Summary

- SOTA Med-MLLMs (MedGemma, Lingshu) scale data and training, which improves medical *language and knowledge* but not what makes an image clinically valuable: the information it adds beyond what the clinical context already predicts.
- **Root issue:** a *bidirectional* image–text information gap. Images contain content reports never state; reports contain content the image cannot support. Under $\max \log p_\theta(T \mid I)$, the second becomes language prior and the first becomes weak perception.
- **Mitigation:** post-training can change how a model *uses* information it already has; it cannot create information earlier stages never kept. Which stage to fix is an empirical question answered by layer-wise probing.
- **Theory:** a four-term loss decomposition (intrinsic, missing context, lost in encoding, not extracted) gives typed uncertainty, a Fano floor on what post-training can fix, and plugs into conformal guarantees.
- **Positioning:** the mechanism is general, but medicine is the structurally extreme case, and general-domain fixes (dense recaptioning) do not transfer.

1 Motivation: why scaling misses the root issue

Scaling improves the model's medical language and knowledge. It does little for *evidential competence*: whether a conclusion is caused by the evidence that justifies it, stated with the confidence that evidence supports, and revised when evidence changes. Four medicine-specific reasons:

1. **Strong prior, tiny signal.** Reports are templated, most studies are normal, and findings occupy a few hundred pixels of a large or 3D volume. Report-text training rewards learning the prior plus the image's overall look.
   - VLMs lose little accuracy on public medical VQA when the image is removed (MIRAGE, as reported in [1]); text-only accuracy on VQA-RAD/SLAKE reaches 11.5–33% [2].
   - In a blinded ED chest X-ray study, Lingshu had a 43.0% rate of clinically significant misses (RADPEER 3b) vs. 13.9% for radiologists, while scoring the best language clarity (88.0% vs. 78.1% for radiologists) [3]. This is exactly what scaling report text optimizes.
   - Our token-zeroing result (concept-grounded cases nearly empty, answers context-grounded) is the mechanistic version: the model uses the image, but not the finding.
   - Costly errors sit where the finding departs from what context predicts (atypical, incidental, rare-in-context cases), a small slice of any benchmark, so aggregate accuracy hides them.
2. **The ground truth is not in the data.** A report is one reader's hedged belief. Truth comes from pathology, follow-up, and outcomes, which are absent from web/PMC corpora (which also

overrepresent publishable cases). In Lingshu's own physician evaluation, two radiologists gave identical scores on only 36% of responses [4]. Scaling reports scales imitation of a noisy reader; "cannot exclude" is learned as style, not belief. MedGemma on real wound images showed heavy, non-actionable hedging [5].

3. **Diagnosis is a process, not a QA pair.** Real reading is indication-guided search, comparison with priors, multi-view integration, and revision when new data arrive. MedBench v5 found stressors mainly disrupt diagnosis updating and contradiction-based self-correction, while final evidence grounding can look superficially stable [6]. MedGemma's critical misses of pericardial effusion and TB without explicit clinical context are the flip side of context dependence [7].
4. **Errors are asymmetric.** A missed pneumothorax counts the same as a wrong nodule size. Clinical value is cost-weighted and lies in knowing when to defer; no current training signal rewards either.

Caveat. Scaling buys knowledge coverage, modality breadth, and instruction-following, which any post-training method needs. The defensible claim is narrower: scaling cannot fix failures caused by the objective and labels rather than by data volume.

2 The root issue: a bidirectional information gap

Let I be the image, T the report, and C the clinical context (indication, history, prior studies, labs). T is written with access to C, but C is usually absent at training time. Report content then falls into four types:

Content	**Example**	**What** $\max p_\theta(T \mid I)$ **teaches**
In image, not in text	Normal anatomy; incidental findings outside the indication	Nothing (no gradient). Perception of unreported content stays weak; reports never say "liver present", consistent with organ-presence failures.
In text, from context	"Unchanged from prior"; "given history of CHF"	Inventing it from priors; report generators fabricate comparisons to non-existent priors [8].
In text, reader-specific	Hedges, template phrasing, judgment calls	Style: "cannot exclude" as habit, not belief.
In both, but predictable from context	"No pneumothorax" on a routine film	Shortcut: the right answer without looking.

Table 1: Decomposition of report content relative to the image.

Refinement 1: mismatch is not the only source of language prior. Rows 2–3 are mismatch. Row 4 is *redundancy*: the text is fully supported by the image, yet the prior gives the same loss more cheaply. The fixes differ: mismatch ⇒ fix the supervision (supply C, strip or tag non-image content); redundancy ⇒ fix the objective (counterfactual pairs where images differ and answers must differ).

Refinement 2: mismatch is a precondition; the objective turns it into failure. Text the image cannot explain still has to be predicted, so the model explains it with the best available predictor, the prior. Language prior is the rational solution to an ill-posed objective: *Med-MLLMs are trained on* $p(T \mid I)$ *when* T *is a function of the image, the context, and the reader.*

Scope. Inter-reader disagreement (partly row 3), prevalence/cost asymmetry, and multi-step revision do not follow from the mismatch. Defensible claim: *hallucination, language prior, and weak perception of unreported findings share one cause.*

Testable prediction. Per-finding error and hallucination rates should track (a) how often the finding is reported when present and (b) how often it is mentioned for non-image reasons; measurable on MIMIC-CXR by tagging report sentences as image-derivable vs. context-derived.

2.1 Figure: the information diagram

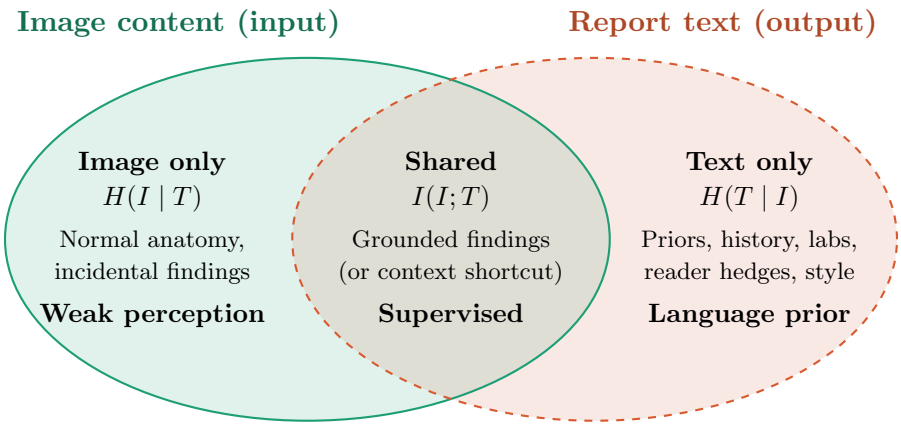


Figure 1: The mutual image–text gap as an information diagram. Ovals are defined over *clinically relevant content*, not raw pixels. The overlap is not uniformly "good": part of it is also predictable from context (the shortcut case); a third circle for C would cut through it.

2.2 The gap ratio varies by data source

Two ratios capture the variation better than one, because sources trade them off:

$$\underbrace{\frac{I(I;T)}{H(T)}}_{\text{text groundedness}} \quad (\text{low} \Rightarrow \text{language prior}), \qquad \underbrace{\frac{I(I;T)}{H(I)}}_{\text{image coverage}} \quad (\text{low} \Rightarrow \text{weak perception}).$$

Implication for the scaling critique. Mixing sources enlarges both ovals, but if new data comes mostly from low-groundedness sources (captions, caption-derived LLM VQA), the overlap *ratio* shrinks, and more data actively trains more language prior: *data volume ≠ grounded supervision.* Test: does a source's groundedness predict the hallucination it contributes?

3 Mitigation: pre-training or post-training?

Rule

Post-training can change how a model *uses* information it already has; it cannot create information that earlier stages never learned to keep. "Pre vs. post" is therefore empirical, and the oracle ladder is the tool that answers it.

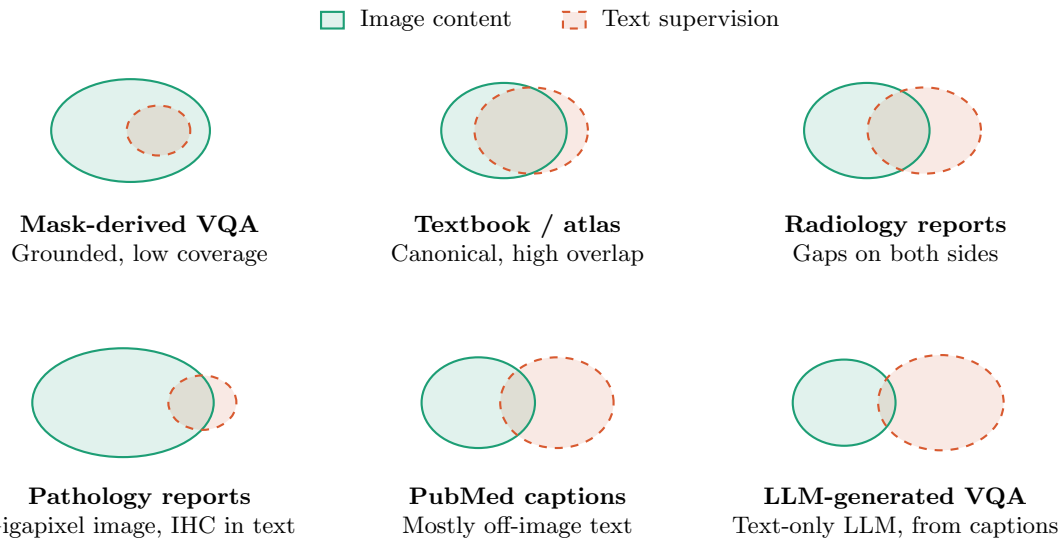


Figure 2: Qualitative (not measured) image–text overlap for six medical data sources, ordered roughly from high to low text groundedness. Measuring these ratios per source is itself a contribution.

Gap (Venn region)	Root cause	Where it is fixable
Coverage (image only)	Unreported content never supervised	Pretraining/alignment, unless the information already survives in the features
Attribution (text only)	Targets the image cannot justify	Pretraining removes the cause; post-training teaches provenance and abstention
Use (overlap, shortcut)	Information present, prior cheaper	Post-training (best target)

Table 2: Which gap can be fixed at which stage.

3.1 Pretraining side (gap-aware)

- **Groundability-aware loss.** Tag report sentences as image-derivable, context-derived, or reader-specific; mask or down-weight non-derivable tokens, or move their source (indication, prior report) into the input so the model learns $p(T \mid I, C)$ instead of inventing C.
- **Coverage expansion.** Dense presence/absence supervision from segmentation models (e.g., organ masks for CT), with a rarity-weighted auxiliary loss (relevance-adaptive alignment).
- **Source-aware mixing.** Weight sources by groundedness, or curriculum from high to low groundedness.

3.2 Post-training side

- **Counterfactual preference optimization.** General-domain versions (image-corrupted preference pairs such as mDPO; visual contrastive decoding) use random corruption, which teaches "the image matters", not "the finding matters". The medical version uses clinically meaningful edits: remove/insert the lesion, swap the prior study, swap the indication, so the preferred answer must change with the finding.
- **Provenance tagging.** Train the model to mark each claim as *seen*, *inferred from context*, or *not determinable from the image*, using a small amount of expert supervision.

3.3 "Uncertainty-aware" is "gap-aware"

The Venn regions map onto uncertainty types: text-only content → missing information; weak overlap → perceptual ambiguity; reader hedges → uncertainty expressed without an epistemic basis. Typed uncertainty thus falls out of the gap frame. Internal-state test: do representations already separate image-grounded from prior-grounded tokens? If yes, steering/ReFT (Trace-and-Steer, Method C) can exploit the separation.

3.4 Deciding experiment and plan

Probe whether gap-relevant findings are decodable at the encoder output, after the projector, and in middle LLM layers of MedGemma/Lingshu. *Decodable but unused* ("can see, doesn't look") ⇒ post-training suffices. *Not decodable* ("can't see") ⇒ pretraining or connector change.

1. **Main method:** post-training on existing backbones (practical; fits CS 598).
2. **Controlled ablation:** standard vs. gap-aware alignment in the LLaVA-style setup, same data.
3. **Paper claim:** *post-training closes the use and attribution gaps; only pretraining closes the coverage gap.* The location of the boundary is itself a finding.

Main risk: surface compliance. The model may learn to hedge more rather than look more. Evaluate on counterfactual pairs (does the answer flip when the lesion is removed?), not on hedging rates or standard benchmarks.

4 Theory for a gap- and uncertainty-robust method

4.1 Loss decomposition along the information path

Let Y be a finding, $Z = g(I)$ the representation the LLM reads, and p_θ the model. The chain rule and the data processing inequality give

$$\mathbb{E}[-\log p_\theta(Y \mid Z)] = \underbrace{H(Y \mid I, C)}_{\text{intrinsic}} + \underbrace{I(Y; C \mid I)}_{\text{missing context}} + \underbrace{I(Y; I \mid Z)}_{\text{lost in encoding}} + \underbrace{\mathbb{E}_Z \,\mathrm{KL}\big(p(Y \mid Z) \,\|\, p_\theta(Y \mid Z)\big)}_{\text{present but not extracted}}. \quad (1)$$

All four terms are nonnegative, and each has a distinct remedy:

Term	Gap	Remedy
Intrinsic (incl. reader disagreement)	–	State it honestly (calibrated)
Missing context	Attribution	Condition on C, or ask for it
Lost in encoding	Coverage	Pretraining / encoder fix only
Not extracted	Use	Post-training

Impossibility (Fano). With frozen Z, any post-trained predictor has error

$$P_e \;\geq\; \frac{H(Y \mid Z) - 1}{\log |\mathcal{Y}|},$$

which turns "post-training cannot fix the coverage gap" into a statement. The math is elementary; the contribution is the framing plus working estimators.

4.2 Estimators

Shannon MI is neither estimable here nor sensitive to usability. Use **$\mathcal{V}$-usable information** [9], which measures information usable by a given predictor family (formalizing "can see but doesn't

look"), and its pointwise form [10]:

$$\mathrm{PVI}_I(y) = \log p(y \mid I) - \log p(y \mid \varnothing).$$

Computed with the image ablated and with context added/removed, PVI estimates the Venn regions per sample and per source, and yields training targets for typed uncertainty without experts. Layer-wise probes (oracle ladder) estimate the lost-in-encoding term. Caveat: all estimates are relative to the probe family.

4.3 Deployment guarantees (conformal)

- **Conformal risk control** [11]: bound the expected miss rate of critical findings at α (handles asymmetric costs).
- **Conformal factuality** [12]: filter report claims so that, w.p. $\geq 1 - \alpha$, every retained claim is correct.
- **Key point:** validity holds for any score; a gap-aware score buys *efficiency* (fewer abstentions at the same guarantee). The decomposition designs the score; a few hundred expert-labeled cases form the calibration set.
- **Limits:** guarantees are marginal; exact per-patient (conditional) coverage is impossible without distributional assumptions [13] (use group-conditional per finding/source). Exchangeability breaks across hospitals; weighted conformal [14] covers covariate shift only.

4.4 Optional tools

Partial information decomposition [15, 16] formalizes the three-circle version (redundancy = shortcut; synergy = "unchanged from prior"). **Group DRO** with groups where the finding contradicts the context prior targets clinically important atypical cases. **Proper scoring rules** [17] as RL rewards make verbalized confidence incentive-compatible.

4.5 Method sketch

- **Train:** post-train with a proper-scoring-rule reward on typed confidence, targets from the PVI-based decomposition.
- **Deploy:** conformal risk control with a gap-aware score.
- **Formal content:** decomposition (proposition), Fano floor (corollary), conformal validity (borrowed).
- **Empirical claim:** gap-aware scores reduce abstention at equal guaranteed risk.

5 Related work and medical specificity

5.1 General-domain work, mapped to the Venn regions

- **Text only.** Concadia [18] separates *descriptions* (visual, meant to replace an image) from *captions* (supply information alongside it) and shows that adding textual context helps both, i.e. the $p(T \mid I, C)$ idea. Text visualness [19] scores sentence-level visualness and maps non-visual text to a NULL image (precedent for groundability tagging). HallE-Switch [20] attributes hallucination to descriptions finer than the vision module can ground; KnowAda [21] adapts captions to the model's visual knowledge and reduces hallucination. Lesson: groundability is relative to the *encoder*, linking the text-only region to the lost-in-encoding term. Text-only analog: fine-tuning on unknown facts encourages hallucination [22].
- **Use gap.** M3ID [23] shows visual reliance decays with generation length and amplifies tokens with high mutual information with the image (decoding-time image PVI). Geigle et al. [24] find

grounding objectives have little to no effect on open-generation hallucination, supporting the view that the problem lies in targets, not alignment capacity.

- **Uncertainty theory.** Kalai et al. [25]: hallucinations are classification errors; benchmarks reward guessing over abstaining; the fix is rewarding calibrated abstention.
- **Naming caution.** Avoid "modality gap" (embedding geometry, Liang et al. 2022); use "information gap" or "supervision gap".

5.2 Why medicine is structurally different

1. Off-image text is *systematic*: radiologists write with priors, history, and labs available by protocol.
2. The image-only region is set by *convention*: exception-based reporting rarely states normal anatomy.
3. The general-domain fix (dense recaptioning: ShareGPT4V, DCI, DOCCI) does not transfer: no reliable teacher for medical images, and dense expert annotation is expensive.
4. Some content is non-visual *in principle* (IHC/molecular status, labs, history).
5. Abstention is a real clinical action (defer, request another view), so Kalai-style incentives fit naturally.

Differentiation from M3ID/VCD. They amplify image dependence uniformly and cannot separate legitimately non-visual content (which should come from C or trigger abstention) from prior-driven hallucination. Amplifying PMI on "unchanged from prior" fixes nothing.

Persuasive experiment. Plot image-PVI distributions for COCO captions, Concadia captions, MIMIC-CXR reports, and PMC captions together. If medical corpora sit far in the text-only regime, "general mechanism, medically extreme" becomes a figure rather than an argument.

6 Next steps

1. Sentence-level groundability tagging of MIMIC-CXR (image-derivable / context-derived / reader-specific); per-source image-PVI and coverage estimates.
2. Layer-wise decodability of gap-relevant findings in MedGemma/Lingshu (oracle ladder) to locate the pre/post boundary.
3. Counterfactual evaluation set (lesion edit, prior swap, indication swap) as the primary metric.
4. Post-training with counterfactual preferences + provenance tagging; conformal calibration with a small expert set.
5. Small-scale gap-aware vs. standard alignment ablation.

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